Read-in action in mixed Hermite space

Two-layer erf nets with a nondiagonal He1–He3 block J

Notes for discussion (notation as in Lipshitz MSc thesis)

Companion to the FCN2 matrix-Laplace / eigenframe calculations in the repo

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Abstract

These notes specialize the thesis notation to a *two-layer* erf network and keep the teacher-aligned Hermite block of the first-layer kernel *nondiagonal*. The thesis mean-field treatment often works in a basis where that block is diagonal; here we keep the off-diagonal matrix element J_{13} explicit. We (i) write the posterior in the mixed He1–He3 mode space, (ii) integrate the readout to obtain a quadratic amplitude matrix G acting on Stein features $c(w)$, (iii) obtain the scalar read-in action $V(w)$ along the teacher axis, and (iv) perform a Ginzburg–Landau expansion that identifies which invariant order parameters control the inner (center) versus outer well transitions, and how ε enters.

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1 Setup and notation (thesis conventions)

We work with a two-layer network with erf activations,

$$f_\Theta(x) = \sum_{i=1}^{N} a_i \, \Phi\big(w_i \cdot x\big), \qquad \Phi = \mathrm{erf}, \tag{1}$$

with read-in rows $w_i \in \mathbb{R}^d$ collected in $W \in \mathbb{R}^{N\times d}$ and readout $a \in \mathbb{R}^N$. (The three-layer objects B, h_k, ν_k of the thesis are absent.)

Data and target. Inputs $x \sim \mathcal{N}(0, I_d)$. The target lives in the span of the first two odd probabilists' Hermite polynomials along a unit teacher direction v_* ($\|v_*\| = 1$):

$$y(x) = \mathrm{He}_1(v_* \cdot x) + \varepsilon\, \mathrm{He}_3(v_* \cdot x), \tag{2}$$

with $\mathrm{He}_1(z) = z$ and $\mathrm{He}_3(z) = \frac{1}{\sqrt{6}}(z^3 - 3z)$ as in the thesis. We write Hermite feature maps $\varphi_1(x) = \mathrm{He}_1(v_* \cdot x)$ and $\varphi_3(x) = \mathrm{He}_3(v_* \cdot x)$.

Inner products. As in the thesis, dataset / population averages use the input measure $d\mu_x$:

$$\langle h \mid g \rangle_x := \int d\mu_x\, h(x)\, g(x), \qquad \|h\|_x^2 := \langle h \mid h \rangle_x. \tag{3}$$

Operator kernels act as $\big(Kg\big)(x) = \int d\mu_{x'}\, K(x, x') g(x')$.

Bayesian action. With noise scale κ (Langevin temperature $T \equiv 2\kappa$) and readout scale χ, the weight-space action has the thesis structure

$$S[a, W] = \frac{N\chi}{2a_0}\, \|a\|^2 + \frac{d}{2s_0}\, \|W\|^2 + \frac{P\chi}{2\kappa}\, \big\|f - y\big\|_x^2, \tag{4}$$

up to inessential overall constants. Here a_0 (code: `sa0`) is the readout prior variance scale and s_0 is the read-in prior variance scale (usually $s_0 = 1$). The posterior is $p(a, W) \propto e^{-S[a,W]}$.

First-layer kernel J. Following the thesis definition of the first-layer activation kernel,

$$J(x, x') := \mathbb{E}_{W \sim p(W|D)} \Big[\tfrac{1}{N} \sum_{i=1}^{N} \Phi(w_i \cdot x)\, \Phi(w_i \cdot x') \Big]. \tag{5}$$

In the large-d Hermite basis aligned with v_*, it is enough for the teacher (2) to retain the 2×2 block of Rayleigh quotients

$$J_{mn} := \langle \varphi_m \mid J \mid \varphi_n \rangle_x, \qquad m, n \in \{1, 3\}, \tag{6}$$

i.e. the finite matrix

$$J = \begin{pmatrix} J_{11} & J_{13} \\ J_{13} & J_{33} \end{pmatrix}. \tag{7}$$

This is the object called Q in the recent FCN2 code notes; here we keep the thesis letter J. The thesis often assumes a basis in which $J_{13} = 0$. We do *not*: mixed space means J_{13} is retained.

Remark 1 (Perp modes)**.** *Modes orthogonal to v_* remain (to leading large-d order) unadapted and are absorbed into an effective floor T in the Stein formulae below; they do not enlarge the 2×2 teacher block.*

2 Posterior in mixed (He1–He3) mode space

Project the network output and the target onto the Hermite subspace $\mathrm{span}\{\varphi_1, \varphi_3\}$. Write mode coefficients

$$f_m := \langle \varphi_m \mid f \rangle_x, \qquad y_m := \langle \varphi_m \mid y \rangle_x, \tag{8}$$

so that, for the target (2),

$$y = \begin{pmatrix} y_1 \\ y_3 \end{pmatrix} = \begin{pmatrix} 1 \\ \varepsilon \end{pmatrix} \tag{9}$$

(up to the conventional normalization of He_3; absolute scales can be absorbed into ε).

In this two-dimensional representation the NNGP / mean-field free action of the thesis (Sec. 2.4.1), restricted to the teacher subspace and with a data-adapted kernel J, reads

$$S_{\mathrm{mix}}[f] = \frac{1}{2}\, f^\top J^{-1} f + \frac{P\chi}{2\kappa}\, \|f - y\|^2, \tag{10}$$

where $f = (f_1, f_3)^\top$ and $\|f-y\|^2 = (f-y)^\top (f-y)$ in mode space. (Equivalently one may keep the population $\langle f \mid J^{-1} \mid f \rangle_x$ and then Rayleigh-project.)

Completing the square / integrating the free field yields the usual ridge resolvent. Define the ridge

$$\rho' := \frac{\kappa}{a_0 P} \quad \text{(effective ridge after the } a_0 \text{ rescaling of } \chi, \kappa \text{ as in the thesis / FCS2)}, \tag{11}$$

and the cavity field

$$v := \big(J + \rho' I_2\big)^{-1} y = \big(J + \rho' I_2\big)^{-1} \begin{pmatrix} 1 \\ \varepsilon \end{pmatrix}. \tag{12}$$

Explicitly, with $\Delta = \det(J + \rho' I)$,

$$v_1 = \frac{1}{\Delta}\big[(J_{33} + \rho') \; - J_{13}\, \varepsilon\big], \tag{13}$$

$$v_3 = \frac{1}{\Delta}\big[-J_{13} + (J_{11} + \rho')\, \varepsilon\big]. \tag{14}$$

Thus **both** cavity components depend on ε and on the mixing element J_{13}. If one set $J_{13} = 0$ (thesis diagonal assumption), these collapse to $v_1 = (J_{11} + \rho')^{-1}$ and $v_3 = \varepsilon (J_{33} + \rho')^{-1}$.

The mode-space posterior mean (learnabilities) is

$$\bar{f} = J\, v = J\big(J + \rho' I\big)^{-1} y, \qquad \tilde{L}_m := \frac{\lambda_m}{\lambda_m + \rho'} \text{ in the eigenbasis of } J. \tag{15}$$

Learnability is therefore controlled by the spectrum of J versus ρ', not by which wells of the eventual read-in potential are occupied.

3 From mixed-space posterior to the amplitude matrix G

Integrating the Gaussian readout a (thesis: integrating the last-layer weights) replaces the data term by a quadratic form in the first-layer features. In the He1–He3 block this produces a 2×2 amplitude matrix of the matrix-Laplace / mean-field form used in the FCN2 calculations,

$$G = \frac{a_0}{N\chi}\Big[-(\chi')^2\, \delta\, v v^\top + \chi'\, J^{-1}\Big], \qquad \chi' := \frac{\chi}{a_0}, \tag{16}$$

with $\delta = 1$ for the standard square loss (discrepancy strength). The first term is negative semidefinite (rank one along v); the second (Onsager) is positive definite when $J \succ 0$.

Remark 2 (Code caveat)**.** *Some experimental solvers temporarily drop the Onsager piece. The structural ε-dependence derived below lives in $-vv^\top$ and is unchanged.*

Writing $G = G_{\mathrm{disc}} + G_{\mathrm{ons}}$,

$$\tfrac{1}{2}\, c^\top G_{\mathrm{disc}}\, c \propto -\,(v_1 c_1 + v_3 c_3)^2, \qquad \tfrac{1}{2}\, c^\top G_{\mathrm{ons}}\, c \propto \chi'\, c^\top J^{-1} c. \tag{17}$$

4 Stein features and the scalar read-in action

Reduce each read-in row to the teacher coordinate $w := w_i \cdot v_*$ (perp components frozen into the thesis floor $T = 1 + 2\frac{d-1}{d} s_0$). The thesis Stein evaluations of $\langle \Phi \circ w \mid \mathrm{He}_n \circ v_* \rangle_x$ define scalar features $c(w) = (c_1(w), c_3(w))$ with

$$c_1(w) = \sqrt{\frac{4}{\pi}} \, \frac{w}{(T + 2w^2)^{1/2}}, \tag{18}$$

$$c_3(w) = -\sqrt{\frac{8}{3\pi}} \, \frac{w^3}{(T + 2w^2)^{3/2}}, \tag{19}$$

normalized so that the empirical / Gaussian-mixture estimates of the block recover

$$J \approx \mathbb{E}_{w \sim p(w)} \big[c(w) c(w)^\top \big] \tag{20}$$

in the Laplace / mixture ansatz for $p(w)$. (We do not assume a sign for c_3 beyond the conventional overall phase in (19); all expansions below keep c_1, c_3 symbolic until the GL section.)

The effective single-neuron read-in action is then

$$\boxed{V(w) = \frac{d}{2s_0} \, w^2 + \tfrac{1}{2} \, c(w)^\top G \, c(w)} \tag{21}$$

with $G = G[J, y, \rho', \chi']$ from (16)–(12). Self-consistency closes by rebuilding J from the wells of V (Laplace mixture: wells $\{\mu_k, \sigma_k\}$ with weights $\pi_k \propto m_k \sigma_k e^{-V(\mu_k)}$, then $J = \sum_k \pi_k \, J(\mu_k, \sigma_k)$).

Where ε enters. Substitute (13)–(14) into the discrepancy square:

$$v_1 c_1 + v_3 c_3 = \underbrace{\big(R_{11}^{-1} c_1 + R_{21}^{-1} c_3 \big)}_{\phi_1(c)} \cdot 1 + \underbrace{\big(R_{12}^{-1} c_1 + R_{22}^{-1} c_3 \big)}_{\phi_\varepsilon(c)} \cdot \varepsilon, \tag{22}$$

where $R := J + \rho' I$. Hence

$$\tfrac{1}{2} c^\top G_{\mathrm{disc}} c \propto -\big(\phi_1(c) + \varepsilon \, \phi_\varepsilon(c) \big)^2 = -\phi_1^2 - \varepsilon^2 \phi_\varepsilon^2 - 2\varepsilon \, \phi_1 \phi_\varepsilon. \tag{23}$$

Only the last term flips under $\varepsilon \to -\varepsilon$. The Onsager piece $\chi' c^\top J^{-1} c$ depends on ε only indirectly through the self-consistent J.

5 Ginzburg–Landau treatment after stripping d

Define the d-stripped potential

$$\tilde{v}(w) := \frac{2s_0}{d} \, V(w) = w^2 + \frac{s_0}{d} \, c(w)^\top G \, c(w). \tag{24}$$

Phase structure of V coincides with that of $\tilde{v}$. The dimensionless coupling is

$$g := \frac{s_0}{d} \times \big(\text{scale of } G \big) \propto \frac{s_0 a_0}{d \, N \chi} \, (\chi')^2. \tag{25}$$

On an invariant training ray one holds the mean-field combinations that keep (g, ρ') fixed while dialling ε (or P).

5.1 Small-w expansion (inner / center transition)

Near $w = 0$,

$$c_1(w) = \alpha_1 w + O(w^3), \qquad c_3(w) = \alpha_3 w^3 + O(w^5), \tag{26}$$

with α_1, α_3 fixed by (18)–(19) and T. Then

$$(v_1 c_1 + v_3 c_3)^2 = (v_1\alpha_1)^2 w^2 + 2v_1 v_3 \alpha_1 \alpha_3 \, w^4 + (v_3\alpha_3)^2 w^6 + \cdots, \tag{27}$$

and (discrepancy-dominated, schematic Onsager absorbed into coefficients)

$$\tilde{v}(w) = a_2 w^2 + a_4 w^4 + a_6 w^6 + \cdots, \tag{28}$$

with invariant Landau coefficients

$$a_2 = 1 - C\,(v_1\alpha_1)^2 + \cdots, \tag{29}$$
$$a_4 = -C\,(2v_1 v_3 \alpha_1 \alpha_3) + \cdots, \tag{30}$$
$$a_6 = -C\,(v_3\alpha_3)^2 + \cdots, \tag{31}$$

and $C \propto g(\chi')^2 \delta > 0$.

Invariant order parameters. Define cavity loads

$$\psi_1 := \sqrt{C}\, v_1, \qquad \psi_3 := \sqrt{C}\, v_3. \tag{32}$$

Then (a_2, a_4, a_6) are functions of the invariants $(\psi_1^2,\ \psi_1\psi_3,\ \psi_3^2)$. From (13)–(14),

$$\psi_1\psi_3 \ \propto\ v_1 v_3 = (\text{terms in } 1) + (\text{terms in } \varepsilon) + (\text{terms in } \varepsilon^2), \tag{33}$$

so $\mathrm{sign}(\varepsilon)$ **moves** a_4 **at fixed** $|\varepsilon|$ through the cross invariant $\psi_1\psi_3$.

Landau diagnostics for the two landscape transitions.

Phenomenon	Condition on (a_2, a_4, a_6)	Controlling invariants
center spinodal	$a_2 = 0$, $a_4 > 0$	ψ_1^2 (mostly He1 load)
inner / central wells	$a_2 > 0$, $a_4 < 0$, $a_6 > 0$	$\psi_1\psi_3 \sim \varepsilon$ cross
outer wells	not visible in small-w; need large-$\lvert w\rvert$ or finite-μ OP	$\lvert\psi_3\rvert$, J_{33}, ρ'
learnability	$\tilde{L} = \lambda(J)/(\lambda + \rho')$	spectrum of J vs ρ'

With $\varepsilon > 0$ and lab mixing, the cross term that feeds a_4 leaves the **inner/center** reorganization live, while the cubic channel $|\psi_3|$ (open outer minima at large $|w|$) provides a **second** landscape transition. Flipping $\varepsilon \to -\varepsilon$ flips only a_4's sensitive piece and can kill the inner transition, leaving a single landscape transition tied to the outer / learning channel.

5.2 Outer transition

The small-w Landau series cannot see outer wells. One either (i) matches the large-$|w|$ asymptotics of $\tilde{v}(w)$, dominated by $-C(v_3 c_3(w))^2$, or (ii) introduces a finite-well order parameter $m = \mu^2$ and expands an effective potential $U(m)$ about a trial outer scale. Criticality is a condition on $(|\psi_3|, \rho', J_{33})$, not on a_2.

5.3 Protocol to map the phase diagram

1. Fix an invariant ray (strip d; hold g and ρ').
2. Solve self-consistent $J \Rightarrow v \Rightarrow (\psi_1, \psi_3)$.
3. Compute (a_2, a_4, a_6) from the small-w expansion of $\tilde{v}$.
4. Draw boundaries in the invariant plane $(\psi_1^2, \psi_1\psi_3)$ or (a_2, a_4).
5. Overlay the outer boundary from a large-w / finite-μ condition.
6. Scan sign(ε) at fixed $|\varepsilon|$: this is a pure a_4 dial if R is held fixed, and diagnoses whether the inner transition is physical or an artifact of the mixing phase.

6 Summary

- Mixed-space posterior: keep the full 2×2 block J including J_{13}; cavity $v = (J + \rho' I)^{-1} y$ with $y = (1, \varepsilon)^\top$.
- Read-in action: $V(w) = \frac{d}{2s_0} w^2 + \frac{1}{2} c(w)^\top G c(w)$ with $G \propto -\chi'^2 \delta\, vv^\top + \chi' J^{-1}$.
- ε enters V through $-\big(\phi_1 + \varepsilon\phi_\varepsilon\big)^2$; only the cross piece $2\varepsilon\phi_1\phi_\varepsilon$ flips with sign(ε).
- After stripping d, Landau coefficients (a_2, a_4, a_6) are controlled by invariants (ψ_1, ψ_3); inner transition $\leftrightarrow a_4 \leftrightarrow \psi_1\psi_3$; outer $\leftrightarrow |\psi_3|$; learning $\leftrightarrow$ spectrum of J vs ρ'.

These notes are intended for discussion with the thesis advisor; they extend the diagonal-J treatment of the MSc thesis to mixed He1–He3 space for the two-layer read-in problem.